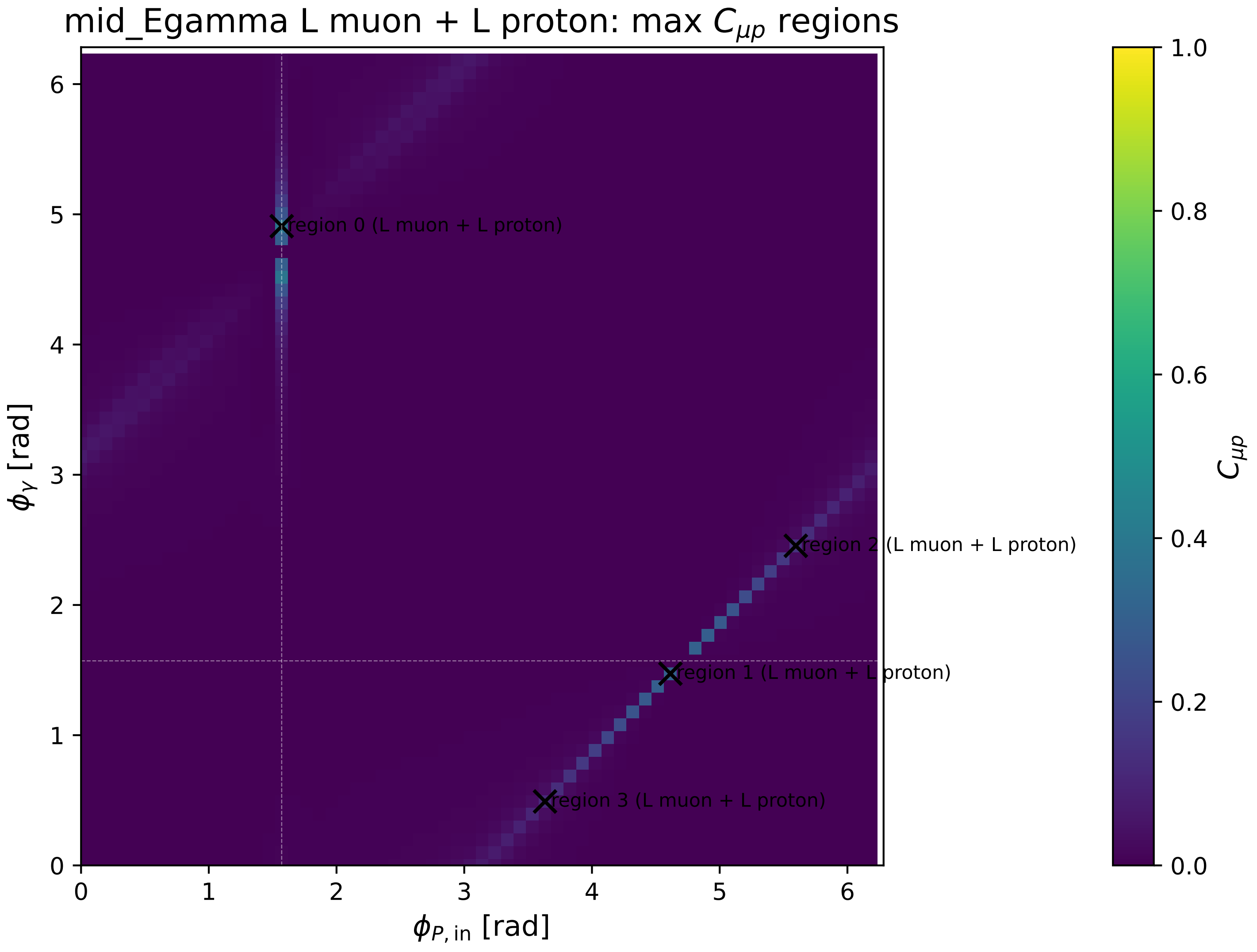
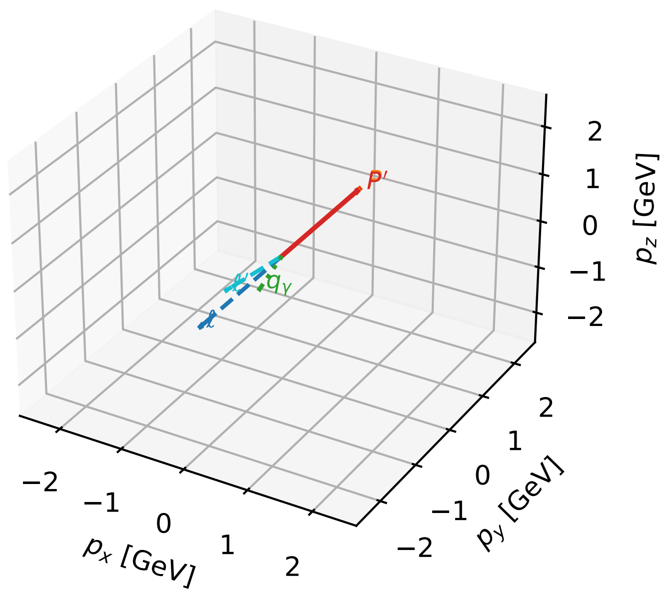


mid_Egamma L muon + L proton: max $C_{\mu p}$ regions
region 0 (L muon + L proton)
region 2 (L muon + L proton)
region 1 (L muon + L proton)
region 3 (L muon + L proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$C_{\mu p}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

L muon + L proton characteristic max $C_{\mu p}$ configuration

3D momenta

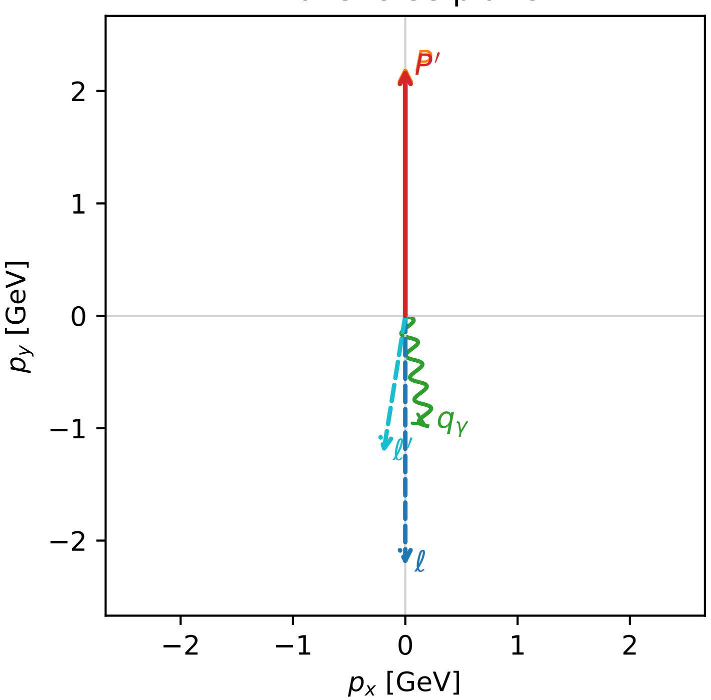


c_mu_p_mid_egamma_ll_region_0 C_{mup}=0.383084
region: mid_Egamma_LL_region_0 final-pair delta_xy=2.98344 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

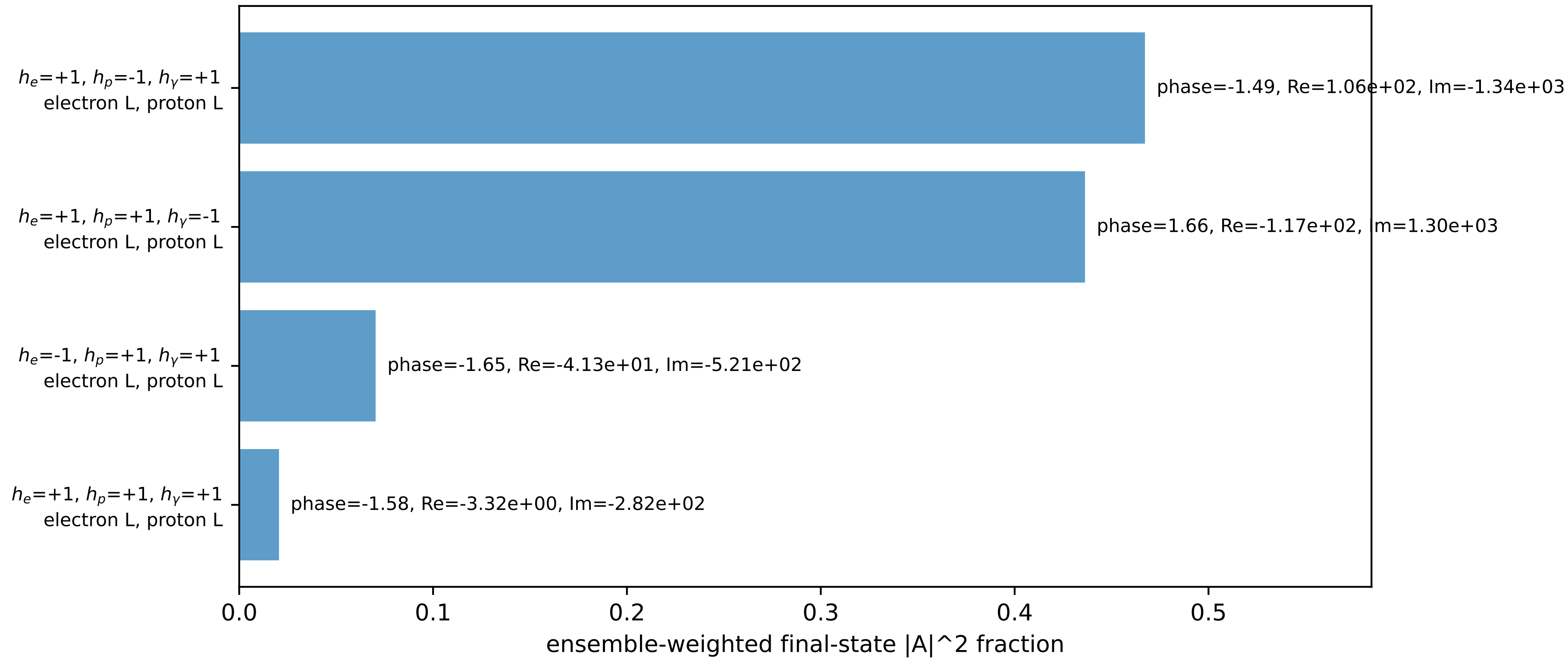
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.20402
θ_{in}=1.57079, ϕ_ℓ=4.71239, ϕ_P=1.5708
E_γ=1, ϕ_γ=4.90874
Q^2=0.0726476, x_B=0.00790796, t=-5.54648e-05, W^2=9.99383, y=0.445416
$\theta(\ell', \gamma)$=20.3116 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= -0.0000 p_y= -2.2231 p_z= -0.0000
P E= 2.4129 p_x= 0.0000 p_y= 2.2231 p_z= 0.0000
ℓ' E= 1.2432 p_x= -0.1951 p_y= -1.2232 p_z= 0.0000
P' E= 2.3953 p_x= 0.0000 p_y= 2.2040 p_z= 0.0000
q_γ E= 1.0000 p_x= 0.1951 p_y= -0.9808 p_z= 0.0000

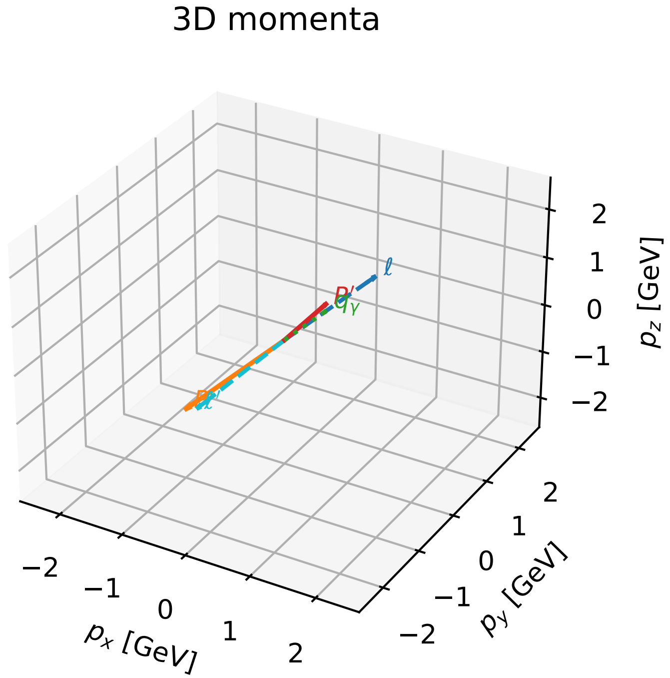
Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=99.4%)



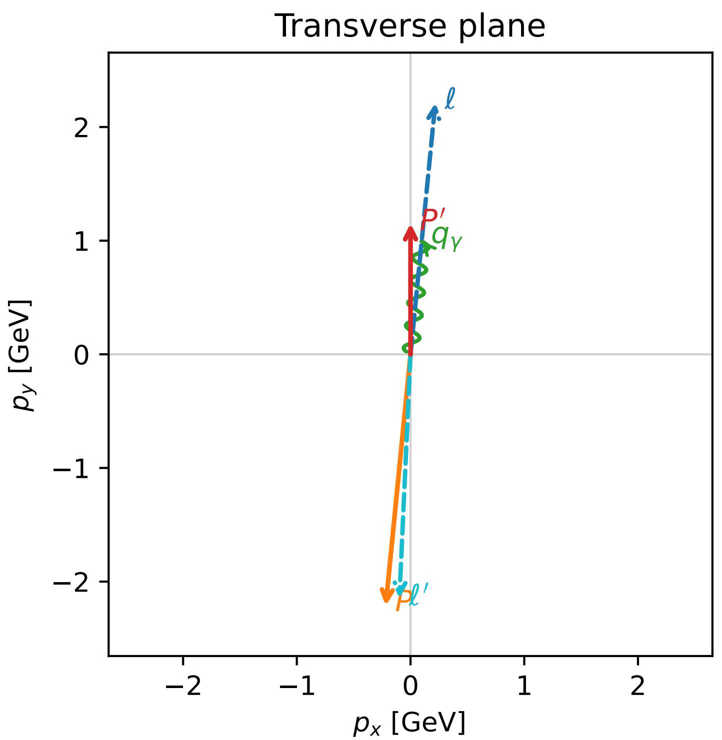
L muon + L proton characteristic max $C_{\mu p}$ configuration

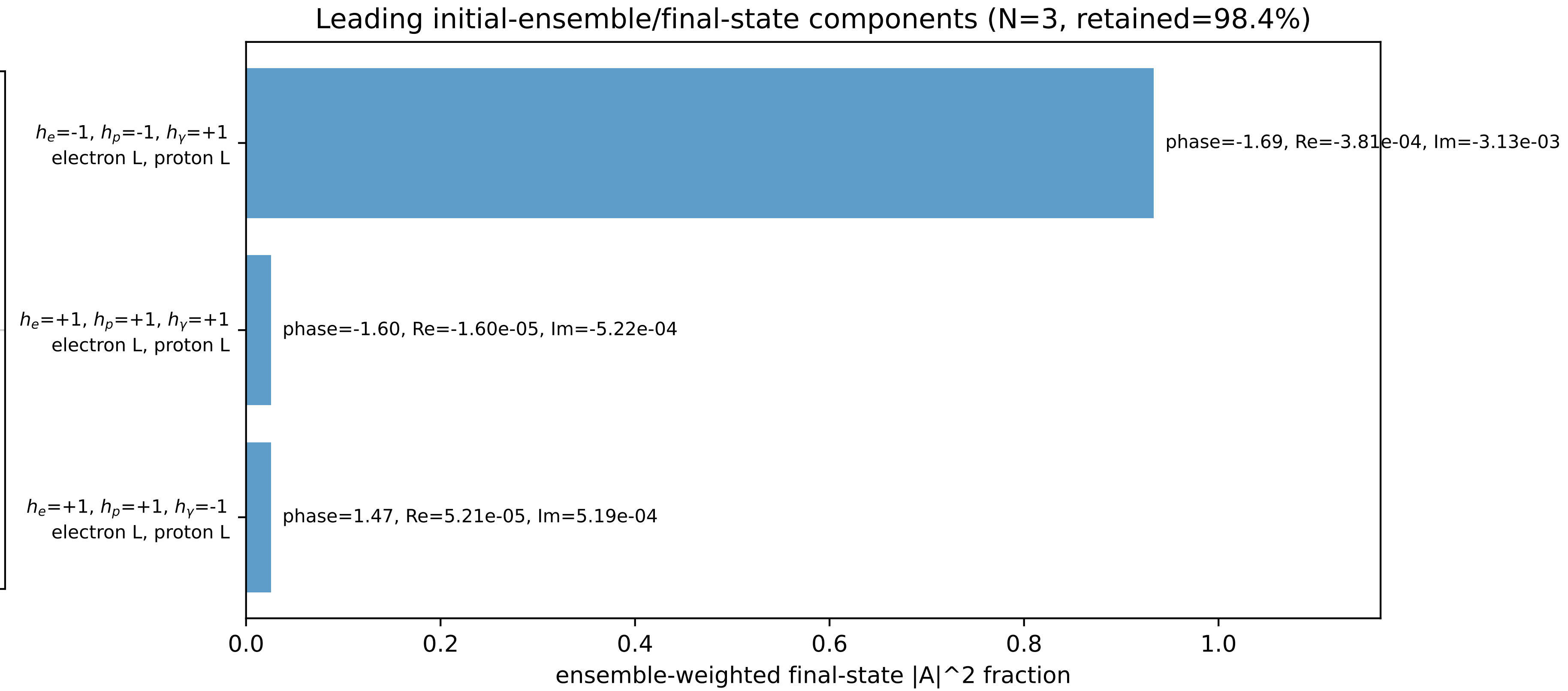



```
c_mu_p_mid_egamma_ll_region_1  C_{mup}=0.307932
region: mid_Egamma_LL_region_1  final-pair delta_xy=3.09599 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
```
incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

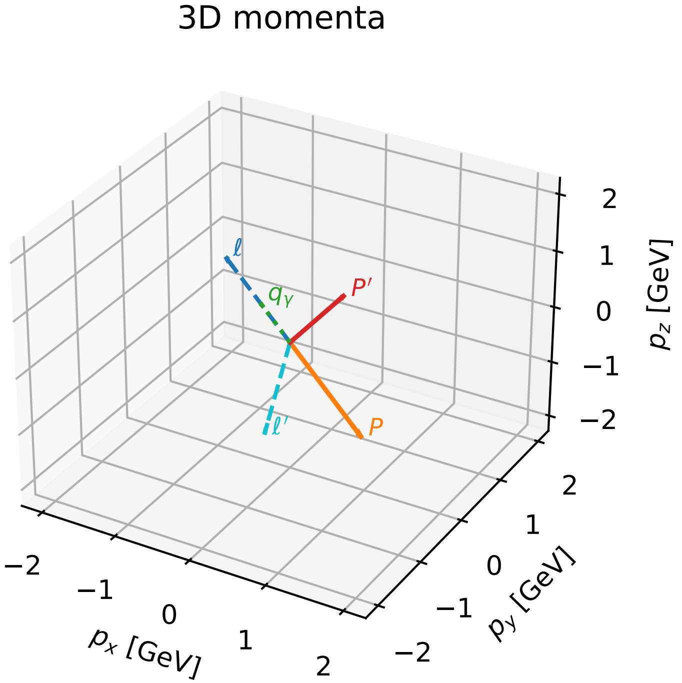
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=1.15252
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.47262, ϕ_P=4.61421
E_γ=1, ϕ_γ=1.47262
Q^2=19.105, x_B=0.965727, t=-10.511, W^2=1.55787, y=0.959186
$\theta(\ell', \gamma)$=176.988 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 0.2179 p_y= 2.2124 p_z= -0.0000
P E= 2.4129 p_x= -0.2179 p_y= -2.2124 p_z= 0.0000
ℓ' E= 2.1525 p_x= -0.0980 p_y= -2.1477 p_z= 0.0000
P' E= 1.4860 p_x= 0.0000 p_y= 1.1525 p_z= 0.0000
q_γ E= 1.0000 p_x= 0.0980 p_y= 0.9952 p_z= 0.0000





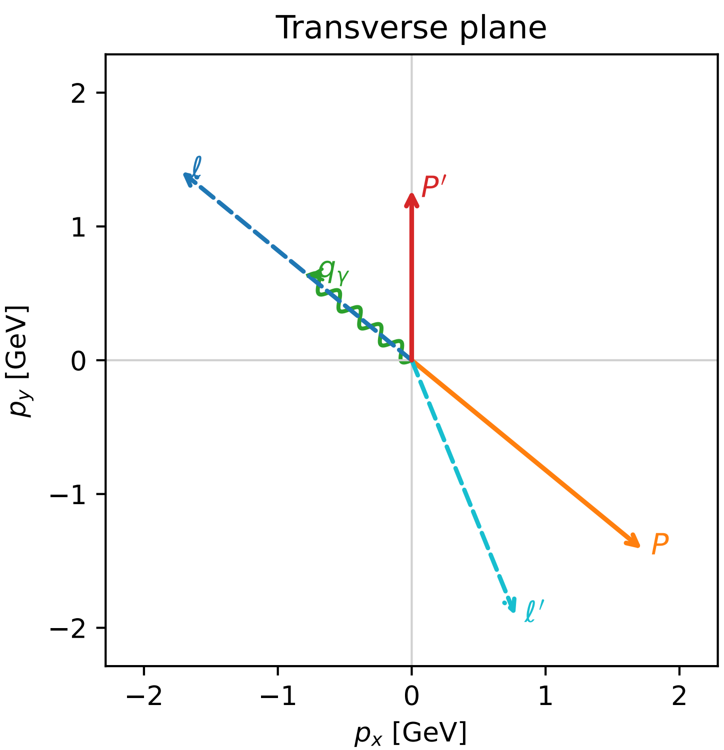
L muon + L proton characteristic max $C_{\mu p}$ configuration

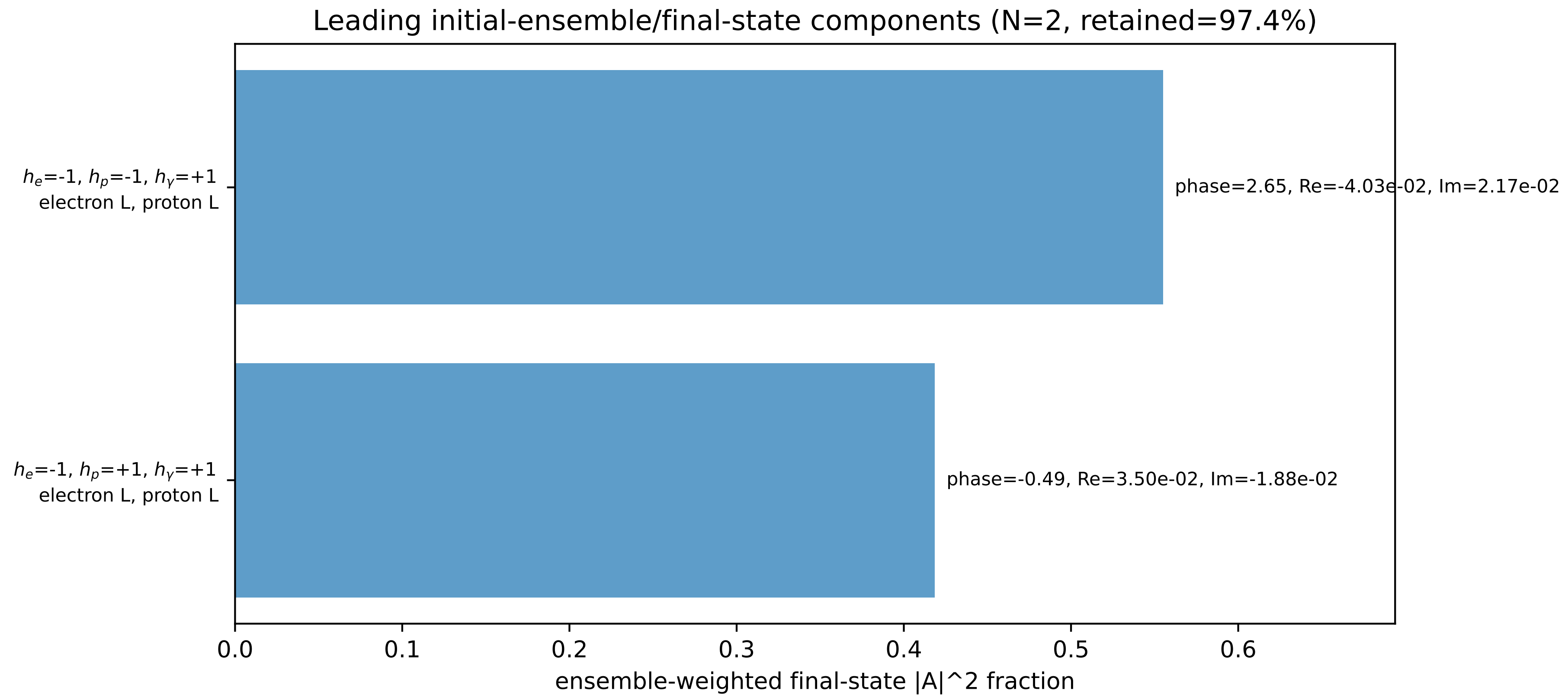


c_mu_p_mid_egamma_ll_region_2 C_{mup}=0.14404
region: mid_Egamma_LL_region_2 final-pair delta_xy=2.75618 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

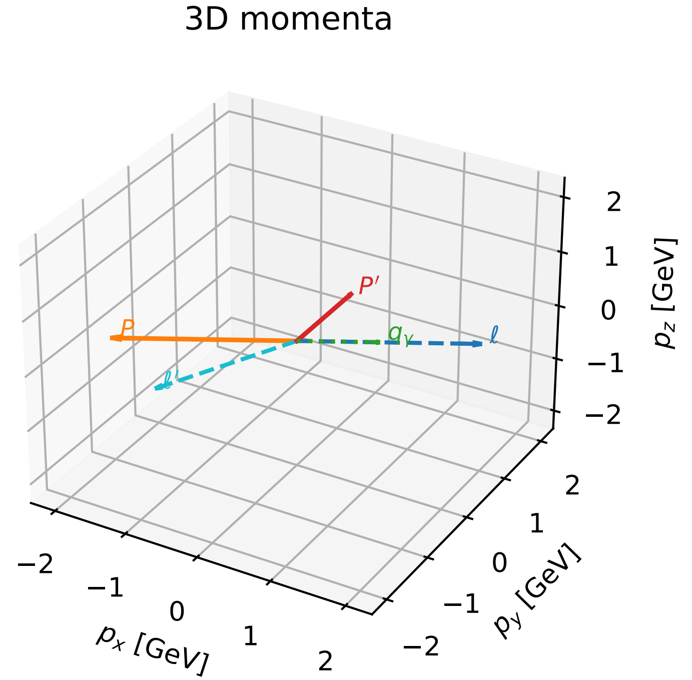
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=1.27096
$\theta_{\rm in}$=1.57079, ϕ_ℓ=2.45437, ϕ_P=5.59596
E_γ=1, ϕ_γ=2.45437
Q^2=17.1735, x_B=0.91738, t=-9.44814, W^2=2.4265, y=0.907651
$\theta(\ell', \gamma)$=151.458 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= -1.7185 p_y= 1.4103 p_z= -0.0000
P E= 2.4129 p_x= 1.7185 p_y= -1.4103 p_z= 0.0000
ℓ' E= 2.0589 p_x= 0.7730 p_y= -1.9054 p_z= 0.0000
P' E= 1.5796 p_x= 0.0000 p_y= 1.2710 p_z= 0.0000
q_γ E= 1.0000 p_x= -0.7730 p_y= 0.6344 p_z= 0.0000





L muon + L proton characteristic max $C_{\mu p}$ configuration



c_mu_p_mid_egamma_ll_region_3 C_{mup}=0.113898
region: mid_Egamma_LL_region_3 final-pair delta_xy=2.68664 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=1.33148
θ_{in}=1.57079, ϕ_ℓ=0.490874, ϕ_P=3.63247
E_γ=1, ϕ_γ=0.490874
Q^2=16.1607, x_B=0.889769, t=-8.89083, W^2=2.88195, y=0.88063
$\theta(\ell', \gamma)$=144.192 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 1.9606 p_y= 1.0480 p_z= -0.0000
P E= 2.4129 p_x= -1.9606 p_y= -1.0480 p_z= 0.0000
ℓ' E= 2.0098 p_x= -0.8819 p_y= -1.8029 p_z= 0.0000
P' E= 1.6287 p_x= 0.0000 p_y= 1.3315 p_z= 0.0000
q_γ E= 1.0000 p_x= 0.8819 p_y= 0.4714 p_z= 0.0000